

Ahmed MOGHAZY

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PROFESSIONAL SUMMARY

Data Science graduate specializing in machine learning, applied AI, and data-driven systems. Experienced in designing and deploying end-to-end ML solutions including AutoML pipelines, predictive modeling, and retrieval-augmented generation (RAG) applications using large language models. Strong background in data analysis, feature engineering, and model evaluation, with hands-on exposure to data engineering workflows, MLOps practices, and translating complex data into actionable business insights.

EDUCATION

Computer Science — *Bachelor of Science*

OCT 2021 - JUNE 2025

Faculty of Computers & AI

GPA: 3.33

Cairo University, Giza, Egypt

Relevant Coursework: Machine Learning, Introduction to Data Science, Mathematical Optimization, Operating Systems, Database Systems, Advanced Databases, Software Engineering, Algorithm Design & Analysis, Object-Oriented Programming

WORK EXPERIENCE

Advanced Computer Technology (ACT) — *Software Developer (AI & Backend)*

OCT 2025 - PRESENT

- Architected a **Multi-Agent Concierge System** using **LangGraph**, orchestrating specialized agents to handle autonomous guest fulfillment and complex policy inquiries with a **95% intent classification accuracy**.
- Engineered a **Hierarchical RAG Pipeline** in **PGVector** to navigate high-density hotel policy documents; implemented recursive retrieval strategies that improved context retention and reduced "hallucinations" by **30%** on long-form queries.
- Integrated LLM **Tool-Access** with **Oracle Opera PMS** via REST APIs, enabling autonomous creation and tracking of guest service requests, reducing manual front-desk entry workload by an estimated **40%**.
- Containerized the entire multi-agent stack using **Docker**, streamlining the CI/CD pipeline and reducing deployment downtime across internal testing environments.

Digital Egypt Pioneers Initiative (DEPI) — *Machine Learning Engineer Trainee*

OCT 2024 - APR 2025

- Completed the Microsoft Machine Learning Engineer track, focusing on designing scalable and production-ready machine learning solutions using Microsoft technologies.
- Built strong foundations in statistics, linear algebra, Python, data preprocessing, visualization, classical ML, deep learning, NLP with attention mechanisms, and computer vision using transfer learning.
- Applied Azure AI Fundamentals and Azure AI Engineer Associate concepts, with hands-on exposure to MLOps tools including MLflow and Hugging Face.

Orange Egypt — *Data Engineer Intern*

AUG 2024 - SEP 2024

- Automated and optimized ETL workflows for company voucher data using Apache NiFi and dbt models.
- Explored pipeline orchestration and scheduling using Apache Airflow to improve data reliability and automation.

ValU — *Data Science Intern*

JULY 2024 - JULY 2024

- Analyzed company growth metrics and customer behavior to generate data-driven strategic insights.
- Designed and delivered interactive Power BI dashboards visualizing KPIs, improving executive decision-making.
- Built and evaluated ML models (Random Forest, XGBoost) for customer lifetime value prediction, achieving an AUC of 0.70.
- Contributed to a customer-facing chatbot by unifying vendors and geographic regions using Named Entity Recognition (NER) for accurate offer retrieval.
- Optimized LLM prompts for intent extraction (offers, vendors, regions) and analyzed OpenAI API usage, reducing inference costs by approximately 30% while maintaining response quality.

Advanced Computer Technology (ACT) — *Data Science Intern*

AUG 2023 - SEP 2023

- Conducted EDA on large datasets and presented insights to stakeholders.
- Automated web scraping pipelines using BeautifulSoup.

TECHNICAL PROJECTS

Project — *AI-Powered Tech News Aggregator (n8n, LLMs)*

JAN 2026 - JAN 2026

- Built an automated news aggregation pipeline using **n8n** to ingest RSS feeds from multiple tech sources on a scheduled basis.
- Implemented an **LLM-based** classification and ranking system to categorize articles, generate *why-it-matters* insights, and score news importance.
- Designed daily and weekly workflows to store structured results in **Google Sheets** and deliver curated HTML email summaries of top-ranked articles.

Graduation Project — <i>AutoML Pipeline (scikit-learn)</i>	OCT 2024 - JUNE 2025
- Engineered a modular AutoML pipeline in Python (scikit-learn) with 4 stages: preprocessing, feature selection, model selection, and hyperparameter optimization. - Synthesized AutoML foundations (model selection, HPO, evaluation) into the system architecture and experiment plan. - Implemented and benchmarked Grid Search, Random Search, and Bayesian Optimization for HPO on tabular ML tasks.	
Project — <i>RAG Chatbot (LangChain + FAISS + Ollama)</i>	DEC 2024 - JAN 2025
- Built a retrieval-augmented chatbot with LangChain + Ollama Gemma-3 (12B) and a Gradio UI for real-time Q&A. - Crawled up to 30 LangChain doc pages (BeautifulSoup); chunked via RecursiveCharacterTextSplitter. - Embedded with all-MiniLM-L6-v2 and indexed in FAISS; boosted relevance using MultiQueryRetriever + Contextual-CompressionRetriever; added SerpAPI + Python REPL tools.	
ExpenSum — <i>Smart Expense Tracker (React + Spring Boot + JWT + LLM)</i>	APR 2025 - MAY 2025
- Developed a full-stack expense tracker: React frontend + Spring Boot backend secured with JWT. - Integrated Mistral (via Ollama) to convert natural-language inputs into structured expense entries for single-step data capture.	
Project — <i>Star-Schema Data Warehouse (SQL, ETL)</i>	DEC 2023 - JAN 2024
- Designed a star schema (fact/dimension) for a recommendation dataset. - Automated daily CSV loads via scheduled SQL jobs (ETL).	

CERTIFICATIONS & ACHIEVEMENTS

DeepLearning.AI — <i>Deep Learning Specialization</i>	OCT 2024 - DEC 2024
Huawei ICT Global Competition — <i>AI Section Lead - Cloud Track</i>	NOV 2024 - MAY 2025
- Won the Second Prize Globally in the Huawei ICT Global Competition (Cloud Track). - Awarded the Grand Prize in the North Africa Region, representing Egypt in the global finals. - Led the AI Technologies section, covering Machine Learning & Deep Learning in the Cloud Track.	

SKILLS

Programming: Python (Advanced), SQL, C++, JavaScript, HTML/CSS

Machine Learning & AI: Scikit-learn, TensorFlow/Keras, Classical Machine Learning, Deep Learning, NLP, Computer Vision, Feature Engineering, Model Evaluation, Hyperparameter Optimization

LLMs & Generative AI: LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Hugging Face, Ollama, FAISS, Vector Databases

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Exploratory Data Analysis (EDA)

Data Engineering & MLOps: Apache Airflow, Apache NiFi, dbt, ETL Pipelines, MLflow, HDFS, Workflow Orchestration

Databases: PostgreSQL, MySQL, Star Schema Design, Data Warehousing

Cloud & Tools: Azure AI Fundamentals, Azure Machine Learning Concepts, Huawei Cloud, Git/GitHub, Docker (Basic), Linux

Core Computer Science: Data Structures & Algorithms, Operating Systems, Distributed Systems, Database Management Systems

Soft Skills: Technical Communication, Problem Solving, Leadership, Analytical Thinking, Adaptability